ARTIFICIAL INTELLIGENCE

CSE 3201

First-Order Logic

Propositional Logic

- Less expressive than predicate logic.
- Is declarative (as opposed to procedural)
- It lacks power to describe things concisely.
- Assumes facts are true or false

First-Order Logic

- Much more powerful than propositional (Boolean) logic
  - Greater expressive power than propositional logic
    - We no longer need a separate rule for each similar sentence
      - Peter is a man → P
      - Paul is a man → Q
      - John is a man → R
    - In PL, can not draw any conclusions about similarities between P, Q, R.

- Sentences like “All men are mortal” can not be represented in Propositional Logic.

- In Predicate Logic (logical extension of Propositional Logic) such sentences can be easily represented.
  - MAN(Peter)
  - MAN(Paul)
  - MAN(John)
  - The limitations of propositional logic are removed to some extent.
  - Similarity between these facts that they are all man can be easily derived.

First-Order Logic

Representing simple facts

It is raining
RAINING

It is sunny
SUNNY

It is windy
WINDY

If it is raining, then it is not sunny
RAINING $\rightarrow \neg$SUNNY

First-Order Logic

- First-Order Logic assumes that the world contains:
  - Objects
    - E.g. people, houses, numbers, John, Dog …
  - Relations
    - E.g. bigger than, comes between, part of, …
  - Functions
    - Some of the relations can be treated as functions. E.g. father of, plus, one more than …

Syntax of First-Order Logic

• Constants	KingJohn, 2, …
• Predicates	Brother, >, …
• Functions	Sqrt, LeftArmOf, …
• Variables	x, y, a, b, …
• Connectives	$\wedge \vee \neg \Rightarrow \Leftrightarrow$
• Equality	$=$
• Quantifiers	$\exists \forall$

Components of First-Order Logic

- Term
  - Constant, e.g. Red
  - Function of constant, e.g. Color(Block1)
- Atomic Sentence
  - Predicate relating objects (no variable)
    - Brother (John, Richard)
    - Married (Mother(John), Father(John))
- Complex Sentences
  - Atomic sentences + logical connectives
    - Brother (John, Richard) $\wedge \neg$ Brother (John, Father(John))

Components of First-Order Logic

- Quantifiers
  - Each quantifier defines a variable for the duration of the following expression, and indicates the truth of the expression…
  - Universal quantifier "for all" ∀
    - The expression is true for every possible value of the variable
  - Existential quantifier "there exists" ∃
    - The expression is true for at least one value of the variable

Truth in First-Order Logic

- Sentences are true with respect to a model and an interpretation
- *"x loves y"* is represented as *LOVE(x, y)* which maps it to true or false when *x* and *y* get instantiated to actual values.
- *"john's father loves john"* is represented as *LOVE(father(john), john)*.
  - Here *father* is a function that maps *john* to his father.
- *x is greater than y* is represented in predicate calculus as *GT(x, y)*.
- It is defined as follows:

$$GT(x, y) = True, \text{ if } x > y$$
$$= False, \text{ otherwise}$$

- Symbols like *GT* and *LOVE* are called predicates
  - Predicates two terms and map to T or F depending upon the values of their terms.

Truth in First-Order Logic

- Translate the sentence *"Every man is mortal"* into Predicate formula.
  - Representation of statement in predicate form
    - "x is a man" and "MAN(x),
    - x is mortal" by MORTAL(x)
  - Every man is mortal :
    -- $(\forall x)\ (MAN(x) \rightarrow MORTAL(x))$

Universal Quantification

- $\forall$ <variables> <sentence>
- $\forall x$ P is true in a model m iff P with x being each possible object in the model
  - Every man is mortal :

    $(\forall x)\ (MAN(x) \rightarrow MORTAL(x))$
- Equivalent to the conjunction of instantiations of P
  - $At(Mike, KSU) \Rightarrow Smart(Mike) \wedge$
  - $At(Laurie, KSU) \Rightarrow Smart(Laurie) \wedge$
  - $At(Sarah, KSU) \Rightarrow Smart(Sarah) \wedge$
  - …

A Common Mistake to Avoid

- Typically $\Rightarrow$ is the main connective with $\forall$
- Common mistake: using $\wedge$ as the main connective with $\forall$
- Typically, $\wedge$ is the main connective with $\exists$
- Common mistake: using $\Rightarrow$ as the main connective with $\exists$

Existential Quantification

- ∃ <variables> <sentence>
- ∃ x P is true in a model m iff P with x being at least one possible object in the model
  - Someone likes McDonalds

    ∃x, likes(x, McDonalds)

- Equivalent to the disjunction of instantiations of P
  - At(Mike, KSU) ∧ Smart(Mike) ∨
  - At(Laurie, KSU) ∧ Smart(Laurie) ∨
  - At(Sarah, KSU) ∧ Smart(Sarah) ∨
  - …

Examples

- Everyone likes McDonalds
  - $\forall x,\ \text{likes}(x, \text{McDonalds})$
- Someone likes McDonalds
  - $\exists x,\ \text{likes}(x, \text{McDonalds})$
- All children like McDonalds
  - $\forall x,\ \text{child}(x) \Rightarrow \text{likes}(x, \text{McDonalds})$
- Everyone likes McDonalds unless they are allergic to it
  - $\forall x, \neg\text{allergic}(x, \text{McDonalds}) \Rightarrow \text{likes}(x, \text{McDonalds})$

Examples

- Every gardener likes the sun.
  -- $\forall x$ gardener(x) $\rightarrow$ likes(x, Sun)
- All purple mushrooms are poisonous.
  -- $\forall x$ (mushroom(x) $\wedge$ purple(x)) $\rightarrow$ poisonous(x)

Properties of Quantifier

- ∀x ∀y is the same as ∀y ∀x
- ∃x ∃y is the same as ∃y ∃x
- ∀x ∃y is not the same as ∃y ∀x
  - **∀x ∃y Loves (x, y)**
    - For everyone (“all x”) there is someone (“exists y”) whom they love.
    - There might be a different y for each x (y is inside the scope of x)
  - **∃y ∀x Loves (x, y)**
    - There is someone (“exists y”) whom everyone loves (“all x”).
    - Every x loves the same y (x is inside the scope of y)

Nesting Quantifier

- There is a kind of food that everyone likes
  $\exists x \ \forall y, \ \text{food}(x) \wedge \text{likes}(y, x)$
- Someone likes all kinds of food
  $\exists y \ \forall x, \ \text{food}(x) \wedge \text{likes}(y, x)$

Examples

- Quantifier Duality
  - Not everyone like McDonalds

    $\neg(\forall x, \text{likes}(x, \text{McDonalds}))$

    $\exists x, \neg \text{likes}(x, \text{McDonalds})$

  - No one likes McDonalds

    $\neg(\exists x, \text{likes}(x, \text{McDonalds}))$

    $\forall x, \neg \text{likes}(x, \text{McDonalds})$

Fun with Sentence

- Brothers are siblings

  $\forall x,y \;\; \text{Brother}(x,y) \Rightarrow \text{Sibling}(x, y)$

- Sibling is “symmetric”

  $\forall x,y \;\; \text{Sibling}(x,y) \Leftrightarrow \text{Sibling}(y, x)$

- One’s mother is one’s female parent

  $\forall x,y \;\; \text{Mother}(x,y) \Leftrightarrow (\text{Female}(x) \wedge \text{Parent}(x,y))$

Other Comments About Quantification

- To say “everyone likes McDonalds”, the following is too broad!
  - $\forall x$, likes(x, McDonalds)
- We mean: Every one (who is a human) likes McDonalds
  - $\forall x$, person(x) $\Rightarrow$ likes(x, McDonalds)
- Essentially, the left side of the rule declares the class of the variable x

Using Predicate Logic

1. Marcus was a man.
2. Marcus was a Pompeian.
3. All Pompeians were Romans.
4. Caesar was a ruler.
5. All Pompeians were either loyal to Caesar or hated him.
6. Every one is loyal to someone.
7. People only try to assassinate rulers they are not loyal to.
8. Marcus tried to assassinate Caesar.

Using Predicate Logic

1. Marcus was a man.

   man(Marcus)

2. Marcus was a Pompeian.

   Pompeian(Marcus)

3. All Pompeians were Romans.

   $\forall x$: Pompeian(x) $\rightarrow$ Roman(x)

Using Predicate Logic

4. Caesar was a ruler.

   ruler(Caesar)

5. All Pompeians were either loyal to Caesar or hated him.

   inclusive-or

   $\forall x$: Pompeian (x) $\rightarrow$ loyalto(x, Caesar) $\vee$ hate(x, Caesar)

   exclusive-or

   $\forall x$: Pompeian (x) $\rightarrow$ (loyalto(x, Caesar) $\wedge \neg$hate(x, Caesar)) $\vee$
   ($\neg$loyalto(x, Caesar) $\wedge$ hate(x, Caesar))

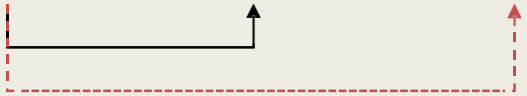
Using Predicate Logic

6. Every one is loyal to someone.

   $\forall x: \exists y:$ loyalto(x, y)

Using Predicate Logic

7. People only try to assassinate rulers they are not loyal to.



$\forall x: \forall y: person(x) \wedge ruler(y) \wedge tryassassinate(x, y)$

$\rightarrow \neg loyalto(x, y)$

Using Predicate Logic

8. Marcus tried to assassinate Caesar.

   tryassassinate(Marcus, Caesar)

Using Predicate Logic

Was Marcus loyal to Caesar?

man(Marcus)

ruler(Caesar)

tryassassinate(Marcus, Caesar)

$\Downarrow$ $\forall x$: man(x) $\rightarrow$ person(x)

$\neg$loyalto(Marcus, Caesar)

Using Predicate Logic

- Many English sentences are ambiguous.
- There is often a choice of how to represent knowledge.
- Obvious information may be necessary for reasoning
- We may not know in advance which statements to deduce (P or ¬P).